

Section : Common

Q.1 Let A be a 3×7 matrix, then each column of A contains:

1. 7 elements
2. 3 elements
3. 21 elements
4. 10 elements

Options

1. 1
2. 2
3. 3
4. 4

Q.2 Match \textbf{List-I} with \textbf{List-II}

List-I

Statement

(A) Degree of the differential equation $\frac{d^2y}{dx^2} = e^{dy/dx}$ is

(B) Order of the differential equation $\left(\frac{dy}{dx}\right)^2 + \frac{d^3y}{dx^3} = 0$ is

(C) Degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 - 5x^2 = 0$

(D) If p is the order and q is the degree of the differential equation $\frac{dy}{dx} + 3y = e^x$, then $p + q$ is

List-II

Its value

(I) 2

(II) not defined

(III) 3

(IV) 1

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)
2. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options

1. 1
2. 2
3. 3
4. 4

Q.3 The solution of the differential equation $(x+1)\frac{dy}{dx} + 1 - 2e^{-y} = 0; y(0) = 0$ is

1. $|(x-1)(e^y - 2)| = 1$
2. $|(x+1)(e^y - 2)| = 1$
3. $|(x+1)(e^y + 2)| = 1$
4. $|(x-2)(e^y - 2)| = 1$

Options

1. 1
2. 2

3. 3

4. 4

Q.4

Value of $\int \left(\frac{1}{\log x} - \frac{1}{(\log x)^2} \right) dx$ is

1. $\frac{x}{\log x} + c$, where c is an arbitrary constant
2. $x \log x + c$, where c is an arbitrary constant
3. $\frac{1}{\log x} + c$, where c is an arbitrary constant
4. $\frac{e^x}{\log x} + c$: where c is an arbitrary constant

Options

1. 1
2. 2
3. 3
4. 4

Q.5

The value of $\int_1^3 \frac{x^2}{x^3 + 1} dx$

1. $\frac{1}{3} \log 26$
2. $\log 14$
3. $\frac{1}{3} \log 14$
4. $\log 56$

Options

1. 1
2. 2
3. 3
4. 4

Q.6

If A is a 3×3 matrix such that $|\text{adj } A| = 9$ and $|kA^{-1}| = 9$, then the value of k are:

1. ± 1
2. $\pm \frac{1}{3}$
3. ± 3
4. $\pm \frac{1}{9}$

Options

1. 1
2. 2
3. 3
4. 4

Q.7

The matrix X in the equation $AX = B$, such that $A = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ is given by

1. $\frac{1}{2} \begin{bmatrix} 2 & -8 \\ 0 & 2 \end{bmatrix}$

2. $\frac{1}{2} \begin{bmatrix} 2 & -6 \\ 0 & 1 \end{bmatrix}$

3. $\frac{1}{2} \begin{bmatrix} 2 & -8 \\ 0 & 1 \end{bmatrix}$

4. $\frac{1}{2} \begin{bmatrix} 1 & -8 \\ 0 & 1 \end{bmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.8 If X is a random variable which can assume values 0, 1, 2, 3 or 4 such that $P(X = 1) = P(X = 2)$ and $3P(X = 3) = 4P(X = 4)$

$= P(X = 0) = \frac{1}{8}$ then $P(X > 0)$ is:

1. $\frac{1}{8}$

2. $\frac{5}{16}$

3. $\frac{19}{96}$

4. $\frac{7}{8}$

Options 1. 1

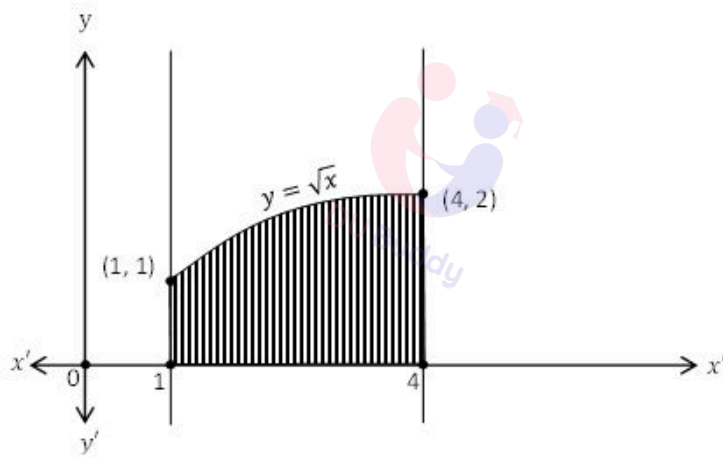
2. 2

3. 3

4. 4

Q.9

The nearest integral value of the shaded area shown below is :



1. 3 square units
2. $\frac{11}{3}$ square units
3. $\frac{14}{3}$ square units
4. 6 square units

Options 1. 1

2. 2
3. 3
4. 4

Q.10

Let A be any skew-symmetric matrix (where A^T is Transpose of matrix A), then which of the following statements are correct?

- (A) A^2 is a symmetric matrix
- (B) A^2 is a skew-symmetric matrix
- (C) $A^T A = -A^2$
- (D) $A^T A - A A^T = O$

Choose the **\textbf{correct}** answer from the options given below:

1. (A) and (B) only
2. (A), (C) and (D) only
3. (B), (C) and (D) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.11 The function $f(x) = 6 - 6x - 2x^2$

1. Strictly decreasing for $x < -\frac{3}{2}$
2. Strictly increasing for $x < \frac{3}{2}$
3. Strictly decreasing for $x > -\frac{3}{2}$
4. Strictly increasing for $x > \frac{3}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.12 The maximum value of $z = 5x + 7y$ subjected to constraints $x + y \leq 5, x \geq 0, y \geq 0$ is:

1. 35
2. 25
3. 40
4. 20

Options 1. 1

2. 2
3. 3
4. 4

Q.13 The function $f(x) = x + \frac{1}{x}$ has

1. local maxima at $x = -1$
2. local minima at $x = -1$
3. local maxima at $x = 1$.
4. neither local maxima nor local minima

Options 1. 1

2. 2
3. 3
4. 4

Q.14 If $x = t^{1/2}, y = t^{3/2}$, then $\frac{dy}{dx} =$

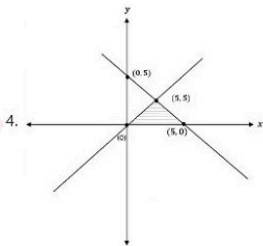
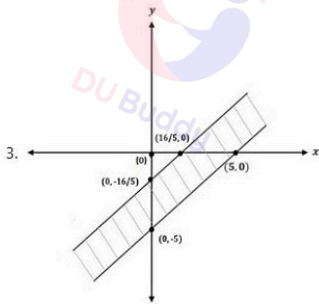
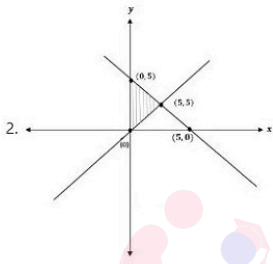
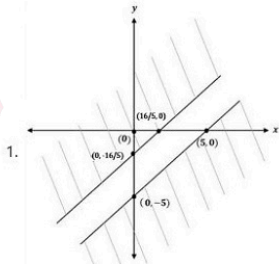
1. $3t$
2. t
3. 3
4. 1

Options 1. 1

2. 2
3. 3

Q.15

Which one of the following represents the correct feasible region determined by the following constraints
 $x - y \geq 5$, $5x - 5y \leq 16$



Options 1. 1

2. 2

3. 3

4. 4

Section : Applied Mathematics

Q.16

For the function, $f(x) = \frac{-3}{4}x^4 - 8x^3 - \frac{45}{2}x^2 - 350$, which of the following statements are correct?

(A) $x = -3$ and $x = -5$ are the only critical points of the given function.

(B) $x = -3$ is a point of local minimum.

(C) The local minimum value at $x = -3$ is 231.

(D) $x = -5$ is a point of local maximum.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (B) and (D) only

3. (C) and (D) only
4. (A), (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.17 The probability distribution function of a normal variate with mean μ and variance σ^2 is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

If $y = f(x)$ be the normal probability curve, then which of the following is correct?

- (A) The normal curve is symmetrical about the line $x = \mu$.
- (B) Mean, median and mode of the distribution coincide.
- (C) Y-axis is an asymptote to the normal curve.
- (D) If x increases numerically, $f(x)$ decreases rapidly.

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Question:

If the matrix $A = \begin{bmatrix} \alpha & \beta & \gamma \\ 0 & 0 & 2 \\ 3 & -2 & 0 \end{bmatrix}$ is a skew symmetric matrix, then the value of $(\alpha + \beta + \gamma)^2$ is:

1. 4
2. 16
3. 9
4. 36

Options 1. 1

2. 2
3. 3
4. 4

Q.19 For the linear programming problem (LPP):

Maximize $Z = x + 1.5y$, subject to constraints,

$x + 2y \leq 40$, $2x + y \leq 40$, $x + y \leq 25$, $x \geq 0$, $y \geq 0$.

Which of the following is NOT correct?

1. The feasible region is bounded.
2. The corner points of the feasible region are (0, 0), (20, 0), (15, 10), (10, 15) and (0, 20).
3. The optimal value of the objective function is attained at the point (15, 10).
4. The LPP has a unique optimal solution.

Options 1. 1

2. 2
3. 3

Q.20 Question:

If a, b, c are positive real numbers, then the least value of

$(a + b + c)(ab + bc + ca)$ is:

1. 1
2. abc
3. $9abc$
4. $3abc$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.21 Question:

The marginal cost (MC) and marginal revenue (MR) functions of a product are

$MC = 20 + \frac{x}{20}$ and $MR = 30$ respectively. If the fixed cost is 200, then

the maximum value of the profit is:

1. Rs. 750
2. Rs. 800
3. Rs. 640
4. Rs. 940

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.22

Which of the following is **not** correct about the Compound Annual Growth Rate (CAGR)?

1. It can be used to demonstrate and compare the performance of investment advisors.
2. It can be used to compare the historical returns of stocks with bonds or with a savings account.
3. It can be used to track the performance of various business measures of one or multiple companies alongside one another.
4. It is calculated by taking the arithmetic mean of series of returns.

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.23

If A and B are two non-singular matrices of order n , then which of the following

statement/statements is/are **not** correct?

- (A) AB is non-singular.
- (B) AB is singular.

(C) $(AB)^{-1} = A^{-1}B^{-1}$

(D) $(AB)^{-1}$ does not exist.

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (A) only
3. (B) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.24 The number of all possible matrices of order 3 with each entry either 0 or 1 is:

1. 18
2. 8
3. 256
4. 512

Options 1. 1

2. 2
3. 3
4. 4

Q.25 A money lender charges Rs 10 for Rs 100 per month in advance then effective

rate of interest per annum charged by money lender is: $\left[\text{given } \left(\frac{10}{9} \right)^{12} = 3.541 \right]$

1. 23.45\%
2. 25.41\%
3. 24.41\%
4. 10\%

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Mr. X wishes to purchase a flat for Rs. 44,65,000 with a down payment of

Rs. 10,00,000 and balance in equated monthly installments (EMI) for 25 years.

If the bank charges 6\% per annum compounded monthly, the EMI is:

[Given: $(1.005)^{300} = 4.4650$]

1. Rs. 25,772

2. Rs. 28,775

3. Rs. 22,325

4. Rs. 30,550

Options 1. 1

2. 2

3. 3

4. 4

Q.27 A die is rolled in such a way that an even number is twice likely to occur as an odd number.

If the die is rolled twice, then the mean of the number of perfect squares in two tosses is:

1. 2

2. $1/3$

3. $2/3$

4. 1

Options 1. 1

2. 2

3. 3

4. 4

Q.28

A boat can row at the speed of 16 km/hr in still water. If the river is flowing at 8 km/hr,

and it takes 8 hours for a round trip, then the distance between the two places is:

1. 36 km

2. 32 km

3. 48 km

4. 54 km

Options 1. 1

2. 2

3. 3

4. 4

Q.29 Which of the following statements are correct about the "Central Limit Theorem"?

(A) The sampling distribution of the sample mean approaches the normal distribution as the sample size gets larger.

(B) A sample size of 30 or more is considered to be sufficient to hold the "Central Limit Theorem".

(C) As the sample size becomes larger, the prediction of characteristics of the population becomes more accurate.

(D) The sampling distribution of the sample mean approaches a bell shaped curve as the sample size gets larger.

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.30

Question: Match the Matrix in List-I with its Adjoint's Determinant in List-II.

List-I (Matrix A):

LIST 1

LIST 2

$$(A) \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$$

9

$$(B) \begin{bmatrix} 5 & -1 \\ 4 & 2 \end{bmatrix}$$

8

$$(C) \begin{bmatrix} 6 & -1 \\ 2 & 1 \end{bmatrix}$$

14

$$(D) \begin{bmatrix} 4 & 1 \\ 3 & 3 \end{bmatrix}$$

2

Choose the correct answer from the options given below:

1. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
2. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
3. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
4. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Options 1. 1

2. 2
3. 3
4. 4

Q.31

The behavior and pattern of the data in a time series is \textbf{NOT} based on which of the following component?

1. Secular trend component
2. Irregular component
3. Cyclic component
4. Stable component

Options 1. 1

2. 2
3. 3

Q.32 It is known that 3\% of plastic bags manufactured in a factory are defective. Using the Poisson distribution on a sample of 100 bags, the probability of at most one defective bag is:

1. $\frac{4}{e^2}$

2. $\frac{4}{e^3}$

3. e^{-3}

4. $\frac{1}{3}$

Options

1. 1

2. 2

3. 3

4. 4

Q.33 Three varieties A, B and C of rice are mixed together in the ratio 1:1:3 respectively. The price of rice A is Rs 127 per kg and that of rice B is Rs 135 per kg. If the price of the mixture is Rs. 152 per kg, then the price per kg of rice of type C is:

1. Rs. 148

2. Rs. 156

3. Rs. 174

4. Rs. 166

Options

1. 1

2. 2

3. 3

4. 4

Q.34 The remainder when $(672 + 541 + 383 + 295 + 101 + 86)$ is divided by 3, is:

1. 0

2. 1

3. 2

4. 3

Options

1. 1

2. 2

3. 3

4. 4

Q.35 If two dice are rolled 12 times and getting a total greater than 4 is considered as a success, then which of the following statements are correct?

(A) The probability of getting a total greater than 4 in a single throw of the pair of dice is $\frac{5}{6}$.

(B) Mean = 10

(C) Variance = $3/5$

(D) The probability of getting a total less than or equal to 4 in a single throw is $1/6$.

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only

2. (A) and (D) only

3. (A), (B), (C) and (D)

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.36 In a time series, the variations which occur due to general tendency of the data to

increase or decrease over a long term are known as:

1. Cyclical Variations

2. Secular Variations

3. Irregular Variations

4. Seasonal Variations

Options 1. 1

2. 2

3. 3

4. 4

Q.37

If A and B are square matrices of the same order, then which of the following statements are correct?

(A) $|A^{-1}| = |A|^{-1}$

(B) $\text{adj}(A) = |A|A^{-1}$

(C) $(A + B)^{-1} = B^{-1} + A^{-1}$

(D) $(AB)^{-1} = B^{-1}A^{-1}$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (A), (C) and (D) only

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.38 In a 500 m race, the ratio of speeds of two participants, A and B, is 4:5 respectively.

If A has a start of 180 m, then the distance by which A wins is:

1. 100 m
2. 120 m
3. 110 m
4. 80 m

Options 1. 1
2. 2
3. 3
4. 4

Q.39 For the given five values, 13, 17, 21, 22, 32; the 3-year moving averages are:

1. 17, 20, 21
2. 17, 21, 25
3. 16, 20, 24
4. 17, 20, 25

Options 1. 1
2. 2
3. 3
4. 4

Q.40 If the following data is obtained from a simple random sample: 6, 7, 9, 10, 11, 17

Then the point estimate of population standard deviation is:

1. 15.21
2. 10
3. 3.898
4. 3.192

Options 1. 1
2. 2
3. 3
4. 4

Q.41

LIST 1 LIST 2

$$\begin{array}{ll} y = \sqrt{x^3} & -1 \\ y = \sqrt{x} & 1 \\ y = x^3 - 47x & 1/4 \\ xy = 16 & 3 \end{array}$$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.42 A motorcycle has a scrap value of Rs. 22,500 after 15 years of its purchase.

If the annual depreciation charge is Rs. 8,500, then the original cost by linear method is:

1. Rs. 1,20,000

2. Rs. 1,50,000

3. Rs. 1,25,000

4. Rs. 1,30,000

Options 1. 1

2. 2

3. 3

4. 4

Q.43

Let $y = 138.86 + 7.64(x - 2021)$ be a straight line of best fit by using least square method.

YEAR(X) 2018 2019 2020 2021 2022 2023 2024

PROFIT(y) 114 130 126 144 138 156 164

Then the trend value for the year 2024 is:

1. 164 2. 154.14 3. 161.78 4. 162.86

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Consider a random sample of 10 students having 116 cm as mean height and standard deviation as 9.798 cm.

If the suggested mean height of the students population is 110 cm, then the t-test statistic of the sample is:

Given: $\frac{\sqrt{10}}{9.798} = 0.3227$

1. 1.93

2. 2.84

3. 1.54

4. 1.64

Options 1. 1

2. 2

3. 3

4. 4

Q.45 About linear programming problem (LPP), which of the following statements are \textbf{correct}?

(A) In a LPP, the linear inequalities or restrictions on the variables are called linear constraints.

(B) If the feasible region for an LPP is unbounded, then the maximum or minimum value of the objective function $Z = ax + by$ never exists.

(C) The feasible region for an LPP is always convex.

(D) The common region determined by all the linear constraints of an LPP is called the feasible region. Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (A), (B), (C) and (D)
4. (B) and (C) only

Options 1. 1
2. 2
3. 3
4. 4

Q.46 Which of the following statements are correct?

(A) A fund which is created to accumulate money over the years to discharge a future obligation is called a sinking fund.

(B) The amount or future value of perpetuity is well-defined.

(C) The sinking fund be used in any emergency.

(D) An equated monthly installment is a fixed payment made by a borrower to a lender at a specific date every month to clear off the loan.

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (D) only
3. (A), (B) and (C) only
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.47 A boat can row at the speed of 16 km/hr in still water. If the river is flowing at 8 km/hr, and it takes 8 hours for a round trip, then the distance between the two places is:

1. 36 km
2. 32 km

3. 48 km

4. 54 km

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Question:

If $x = t^3$ and $y = t^2$, then $\frac{d^2y}{dx^2}$ is equal to:

1. $\frac{-1}{t^2}$ 2. $\frac{1}{2t^3}$ 3. $\frac{-1}{t^3}$ 4. $\frac{-2}{9t^4}$

Options 1. 1

2. 2

3. 3

4. 4

Q.49 At what rate of interest will the present value of a perpetuity of Rs. 1000

payable at the end of every six months be Rs. 40000?

1. 10\% 2. 5\% 3. 7\% 4. 12\%

Options 1. 1

2. 2

3. 3

4. 4

Q.50

The value of the definite integral $\int_0^1 e^x \frac{(1-x)^2}{(1+x^2)^2} dx$ is:

1. $\frac{e}{2}$ 2. $\frac{e}{2} - 1$ 3. $\frac{e}{2} + 1$ 4. $2e + 1$

Options 1. 1

2. 2

3. 3

4. 4

Section : Core Mathematics

Q.51

$\int_0^{\frac{\pi}{2}} \sqrt{1 - \sin 2x} dx$ is equal to:

1. $2(\sqrt{2} - 1)$

2. $2(\sqrt{2} + 1)$

3. 2

4. $2\sqrt{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.52

The vector in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$ that has magnitude 9 units is:

1. $2\hat{i} - \hat{j} - 2\hat{k}$

2. $9(2\hat{i} - \hat{j} - 2\hat{k})$

3. $\frac{1}{9}(2\hat{i} - \hat{j} - 2\hat{k})$

4. $(6\hat{i} - 3\hat{j} - 6\hat{k})$

Options 1. 1

2. 2

3. 3

4. 4

Q.53 A relation $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(x) = x^2, x \in \mathbb{N}$ (Set of Natural numbers) , Then $f(x)$ is

1. neither injective nor surjective

2. injective only

3. surjective only

4. bijective

Options 1. 1

2. 2

3. 3

4. 4

Q.54 If $\begin{vmatrix} 1 & \cos \theta & 0 \\ \sin \theta & 1 & \cos \theta \\ \cos \theta & 1 & -\sin \theta \end{vmatrix} = A \sin \theta + B \cos \theta + C \sin \theta \cos \theta$ then:

1. $A + B = 0$

2. $B + C = 2$

3. $C + A = 1$

4. $B - A = 1$

Options 1. 1

2. 2

3. 3

4. 4

Q.55 If $y = \sin^{-1} \sqrt{\frac{x}{x+1}} + \sec^{-1} \sqrt{\frac{x+1}{x}}$, then $\frac{dy}{dx}$ is

1. 0

2. 1

3. $\frac{1}{\sqrt{x+1}}$

4. $\frac{1}{\sqrt{x}}$

Options 1. 1

2. 2

3. 3

4. 4

Q.56 Let $R = \{(L_1, L_2) : L_1 \perp L_2 \text{ where } L_1, L_2 \in L \text{ (set of straight line in a plane)}, \text{ then}$

1. R is reflexive but neither symmetric nor transitive
2. R is symmetric but neither reflexive nor transitive
3. R is an equivalence relation
4. R is reflexive and transitive but not symmetric

Options 1. 1

2. 2

3. 3

4. 4

Q.57 Differentiation of $\log[\log(\log x^5)]$ with respect to x is

1. $\frac{5}{x(\log x^5) \log(\log x^5)}$

2. $\frac{5}{x \log(\log x^5)}$

3. $\frac{5x^2}{(\log x^5) \log(\log x^5)}$

4. $\frac{5x^4}{(\log x^5) \log(\log x^5)}$

Options 1. 1

2. 2

3. 3

4. 4

Q.58 If $\vec{a}$ and $\vec{b}$ are two non-zero vectors such that $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$, then the angle θ between $\vec{a}$ and $\vec{b}$ is

1. $\frac{\pi}{2}$

2. $\frac{\pi}{4}$

3. $\frac{\pi}{6}$

4. $\frac{\pi}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.59 Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I}

\textbf{List-II}

(A) Integrating factor of $xdy - (y + x^2)dx = 0$ (I) x^2

(B) Integrating factor of $xdy + (2y + x^2)dx = 0$ (II) x^3

(C) Integrating factor of $(3y - x^2)dx + xdy = 0$ (III) x

(D) Integrating factor of $(y + 3x^2)dx + xdy = 0$ (IV) $\frac{1}{x}$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.60

If $\int \frac{x^4}{x-2} dx = px + qx^2 + rx^3 + sx^4 + t \log|x-2| + C$, where C is an arbitrary constant and p, q, r, s, t are real numbers,

then the correct arrangement of p, q, r, s, t is

1. $p > q > r > s > t$

2. $q > r > p > t > s$

3. $r > t > s > p > q$

4. $t > p > q > r > s$

Options 1. 1

2. 2

3. 3

4. 4

Q.61

The area of triangle with vertices P, Q, R is given by (where $\vec{AB}$ = position vector of point B - position vector of point A)

1. $\frac{1}{8} |\vec{PR} \times \vec{PQ}|$

2. $\frac{1}{4} |\vec{PR} \times \vec{PQ}|$

3. $\frac{1}{2} |\vec{PQ} \times \vec{PR}|$

4. $|\vec{PQ} \times \vec{PR}|$

Options 1. 1

2. 2

3. 3

4. 4

Q.62

The shortest distance between the following lines:

$$\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + s(2\hat{i} + \hat{j} + \hat{k})$$

$$\vec{r} = (\hat{i} + \hat{j} + 2\hat{k}) + t(4\hat{i} + 2\hat{j} + 2\hat{k}), \text{ where } s \text{ and } t \text{ are scalars, is:}$$

1. $\frac{3\sqrt{6}}{\sqrt{5}}$

2. $\sqrt{\frac{15}{2}}$

3. $\frac{\sqrt{5}}{3\sqrt{6}}$

4. $\sqrt{\frac{2}{15}}$

Options

1. 1
2. 2
3. 3
4. 4

Q.63 Let A and B be independent events such that $P(A) = 0.3$ and $P(B) = 0.4$, then

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \textbf{List-II}

(A) $P(A \cap B)$ (I) 0.3

(B) $P(A \cup B)$ (II) 0.4

(C) $P(A|B)$ (III) 0.12

(D) $P(B|A)$ (IV) 0.58

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

4. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.64 The system of linear equations $kx + 5y = 5$, $2x + 3y = 5$ will be consistent if

1. $k \neq \frac{5}{3}$

2. $k \neq 5$

3. $k \neq \frac{10}{3}$

4. $k = \frac{10}{3}$

Options

1. 1
2. 2
3. 3
4. 4

Q.65 The angle between the line $2x = 3y = z$ and x - axis is:

1. $\cos^{-1}\left(\frac{1}{3}\right)$

2. $\cos^{-1}\left(\frac{3}{7}\right)$

3. $\cos^{-1}\left(\frac{2}{7}\right)$

4. $\cos^{-1}\left(\frac{1}{7}\right)$

Options 1. 1

2. 2

3. 3

4. 4

Q.66 A letter is known to have come from either TATAPUR or from CHAKRATA. On the envelope, only two letters "TA" are visible consecutively. The probability that the letter has come from CHAKRATA is:

1. $\frac{1}{10}$

2. $\frac{3}{10}$

3. $\frac{1}{7}$

4. $\frac{1}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.67 $\sin^{-1}\left(\cos \frac{3\pi}{5}\right)$ equals

1. $\frac{3\pi}{5}$

2. $-\frac{\pi}{10}$

3. $\frac{\pi}{10}$

4. $-\frac{3\pi}{10}$

Options 1. 1

2. 2

3. 3

4. 4

Q.68 If $A = \begin{bmatrix} 4 & 3 \\ 2 & -1 \\ 1 & 0 \end{bmatrix}$ and $B^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -1 \end{bmatrix}$, then $A - B$ is equal to

1. $\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & -1 \end{bmatrix}$

2. $\begin{bmatrix} 3 & 1 \\ -2 & 0 \\ -2 & 1 \end{bmatrix}$

3. $\begin{bmatrix} 3 & 1 \\ 0 & -2 \\ -2 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 4 & 3 \\ 2 & -1 \\ 1 & 0 \end{bmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.69

The solution of the differential equation $\frac{dy}{dx} = \frac{ax + c}{by + d}$ represents a circle when

1. $a = -b$

2. $a = b$

3. $a = -2b$

4. $a = 3b$

Options 1. 1

2. 2

3. 3

4. 4

Q.70

Area (in sq. units) of the region bounded by curves $y^2 = x$ and $x = 4$ is

1. $\frac{16}{3}$

2. $\frac{32}{3}$

3. $\frac{8}{3}$

4. $\frac{64}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.71

If x, y and z are non-zero distinct numbers, then

$$\begin{vmatrix} x+y & y+z & z+x \\ z & x & y \\ 1 & 1 & 1 \end{vmatrix} \text{ is equal to}$$

1. $x + y + z$

2. $2(x + y + z)$

3. 1

4. 0

Options 1. 1

2. 2

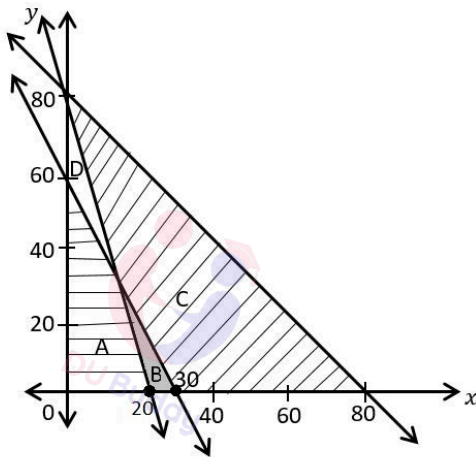
3. 3

4. 4

Q.72

Which of the region shown in the given figures represents the feasible region bounded by the following constraints?

$$4x + y \geq 80, 2x + y \geq 60, x + y \leq 80, x \geq 0, y \geq 0$$



1. region A
2. region B
3. region C
4. region D

Options 1. 1

2. 2

3. 3

4. 4

Q.73

If lines $\frac{x+5}{5\lambda+2} = \frac{4-2y}{10} = \frac{1-3z}{-3}$ and $\frac{x-2}{1} = \frac{1+2y}{4\lambda} = \frac{2+z}{3}$ are perpendicular, then value of ' λ ' is

1. - 1

2. 1

3. 2

4. - 2

Options 1. 1

2. 2

3. 3

4. 4

Q.74

If the area of a triangle with vertices $(-3, 0)$, $(3, 0)$ and $(0, k)$ is 9 sq. units, then k equals

1. 0

2. - 3

3. ± 3

4. 3

Options 1. 1

2. 2

3. 3

4. 4

Q.75

If $\vec{a} = 2\hat{j} - \hat{k}$, $\vec{b} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{c} = -\hat{i} + \hat{k}$ are three vectors, then the area (in sq. units) of the parallelogram whose diagonals are $(\vec{b} + \vec{c})$ and $(\vec{a} + \vec{c})$ is

1. $\sqrt{21}$

2. $\frac{1}{2}\sqrt{21}$

3. $\sqrt{42}$

4. 21

Options 1. 1

2. 2

3. 3

4. 4

Q.76

Interval in which the function f given by $f(x) = \tan x - 4x$, $x \in \left(0, \frac{\pi}{2}\right)$ is strictly decreasing is

1. $0 < x < \frac{\pi}{3}$

2. $\frac{\pi}{3} < x < \frac{\pi}{2}$

3. $\frac{\pi}{4} < x < \frac{\pi}{2}$

4. $\frac{\pi}{2} > x > 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.77

For the function $f(x) = 2x^3 - 3x^2 - 12x + 5$, the difference of maximum and minimum value of $f(x)$ is

1. 12

2. -15

3. 27

4. -3

Options 1. 1

2. 2

3. 3

4. 4

Q.78

If the maximum value of the function $f(x) = \frac{2 \log_e x}{x}$, $x > 0$ occurs at $x = e$, then $e^3 f''(e)$ is equal to

1. -2

2. $-\frac{2}{e}$

3. $-e$

4. $-5e^3$

Options

1. 1
2. 2
3. 3
4. 4

Q.79

If $f(x) = \begin{cases} mx + 1, & x \geq \pi/2 \\ \sin x + n, & x \leq \pi/2 \end{cases}$ is continuous at $x = \pi/2$, where $m \in Z$ (set of integers), then $\sin 2n =$

1. 0

2. 1

3. -1

4. a finite value between -1 and 0

Options

1. 1
2. 2
3. 3
4. 4

Q.80

The probability that it will rain on any particular day is 50%. The probability that it rains only on the first 4 days of the week is:

1. $\frac{1}{64}$

2. $\frac{1}{128}$

3. $\frac{1}{16}$

4. $\frac{1}{8}$

Options

1. 1
2. 2
3. 3
4. 4

Q.81

The corner points of the bounded feasible region determined by the system of linear constraints are $(0, 0), (5, 0), (6, 5), (6, 8), (4, 10), (0, 8)$. Let $Z = 3x - 4y$ be the objective function. The minimum value of Z occurs at

1. $(0, 0)$

2. $(5, 0)$

3. $(0, 8)$

4. $(4, 10)$

Options

1. 1
2. 2
3. 3

4. 4

Q.82

If $\begin{bmatrix} x-y & 0 \\ x+y & 1 \end{bmatrix}$ is an identity matrix and $\begin{bmatrix} x & y \\ z & x \end{bmatrix}$ is a singular matrix then:

1. $x > y = z$
2. $x = y > z$
3. $x = y < z$
4. $x \neq y \neq z$

Options 1. 1

2. 2
3. 3
4. 4

Q.83

Value of $\int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} dx$ is

1. 0
2. $\frac{1}{2}$
3. 1
4. 5

Options 1. 1

2. 2
3. 3
4. 4

Q.84

A couple has 3 children each child is equally likely to be a boy or a girl. The probability that the eldest child is a girl given that they have atleast one boy is:

1. $\frac{1}{2}$
2. $\frac{3}{7}$
3. $\frac{4}{7}$
4. $\frac{3}{8}$

Options 1. 1

2. 2
3. 3
4. 4

Q.85

$\int \frac{\sin x - x \cos x}{x(x + \sin x)} dx =$ (where C is an arbitrary constant)

1. $\log \left| \frac{x}{x - \sin x} \right| + C$
2. $\log \left| \frac{x}{x + x \sin x} \right| + C$

3. $\log \left| \frac{x}{x + \sin x} \right| + C$

4. $\log \left| \frac{x + \sin x}{x} \right| + C$

Options

1. 1

2. 2

3. 3

4. 4